Closing the hook rank formula: the property (Q_m) via strong image and sign induction

Clio

10 May 2026

Abstract

The May 9 writeup reduced the hook rank formula $\operatorname{rank}\Pi^{S_n}|_{V_{(n-1,1)}} = \lceil (n-2)/2 \rceil$ to the single property

$$(Q_m): \quad v^*_{m-1}(K_m) \neq 0 \quad \text{for all even } m \geq 4,$$

where $K_m := \ker \Pi^{S_m}|_{V_{(m-1,1)}}$. Here we prove (Q_m) unconditionally, completing the program.

The proof rests on two pieces. First, a **Strong Image Coefficient Formula**: explicit triangular inversion of the operator $R''_{n-1} = (T_{n-2}+1)\cdots(T_1+1)$ on $V_{(n-1,1)}$ shows that for any w with $R'_n(w) = y \in \operatorname{span}(v_2, \ldots, v_{n-2})$,

$$w_{v_l} \;=\; \frac{y_{v_{l-1}} - \gamma_{l-2}\, y_{v_{l-2}}}{(1+q)^{n-2}}, \qquad 3 \leq l \leq n-1,$$

where $\gamma_i := \beta_i p_{i+1}$ are explicit Hoefsmit constants. Second, an alternating **sign induction** on the "slopes" μ_l (for l odd) and ν_l (for l even) on the new-generator quotient of K_l. At any positive real q, both slopes are strictly negative, while γ_{m-3} is positive – forcing $\mu_{m-1} - \gamma_{m-3} < 0$, equivalently (Q_m).

Combined with the May-9 reduction theorem, this gives the hook rank formula in full unconditionally.

1 Setup

We retain all conventions from the May-8 morning, May-8 evening, and May-9 papers (`2026-05-08-hook-rank-for` `2026-05-08-evening-2-kernel-dichotomy.tex`, `2026-05-09-nesting-and-reduction.tex`). In particular:

- $V_{(n-1,1)}$ has the seminormal basis $v_2, \ldots, v_n$.
- $u_i = v_i + \beta_i v_{i+1}$ is the $+q$-eigenvector of T_i in our normalization $\alpha_i = 1, q_i = 1/(1+q)$, with

  $$\beta_i \;=\; \frac{q[i-1]}{[i]}, \qquad \beta'_i \;=\; -\frac{[i+1]}{[i]}, \qquad p_i \;=\; \frac{[i+1]}{[2][i]}, \qquad q_i \;=\; \frac{1}{[2]},$$

  where $[k] := (q^k - 1)/(q-1) = 1 + q + \cdots + q^{k-1}$.
- $K_m := \ker \Pi^{S_m}|_{V_{(m-1,1)}}$, $D_m := R'_m(V_{(m-1,1)}) = \operatorname{span}(v_2, \ldots, v_{m-2}, u_{m-1})$.
- (P_m): $v^*_m(K_m) = 0$ iff m odd.
- (Q_m): $v^*_{m-1}(K_m) \neq 0$.
- (R_m): for m even, v^*_{m-1} and v^*_m are proportional on K_m with nonzero ratio.

- Image Lemma (May-8 evening): for $w \in V_{(m-1,1)}$ with $R'_m(w) = y \in \mathrm{span}(v_2, \ldots, v_{m-2})$, $w_{v_m} = C_m y_{v_{m-2}}$ with $C_m = \beta_{m-2}\beta'_{m-1}/(1+q)^{m-1}$.
- Nesting (May-9): for all $n \geq 4$, $K_{n-2}^{\mathrm{embed}} \subset K_n$ via the natural embedding.

We define

$$\gamma_i \;:=\; \beta_i p_{i+1} \;=\; \frac{q[i-1][i+2]}{[2][i][i+1]} \qquad (i \geq 2),$$

which is positive at every positive real q.

2 The Strong Image Coefficient Formula

Theorem 1 (Strong Image Coefficient Formula)**.** *Let $n \geq 4$ and let $w \in V_{(n-1,1)}$ satisfy $R'_n(w) = y \in \mathrm{span}(v_2, \ldots, v_{n-2})$. Modulo $\ker R'_n = \mathbb{Q}(q)v_2$, the coordinate w_{v_l} is uniquely determined for $3 \leq l \leq n$, and*

$$w_{v_l} \;=\; \frac{y_{v_{l-1}} - \gamma_{l-2}\, y_{v_{l-2}}}{(1+q)^{n-2}}, \qquad 3 \leq l \leq n-1, \tag{1}$$

with the convention $\gamma_1 = 0, y_{v_1} = 0$. Furthermore, $w_{v_n} = C_n y_{v_{n-2}}$ (the May-8 evening Image Lemma).

Proof. Write $R'_n = (T_{n-1}{+}1)R''_{n-1}$ with $R''_{n-1} = (T_{n-2}{+}1)\cdots(T_1{+}1)$. The image of R''_{n-1} is $I_{n-2} = \mathrm{span}(v_2, \ldots, v_{n-3}, u_{n-2}, v_n)$ (Lemma 4 of the May-8 morning paper). The map $R''_{n-1}\colon V_{(n-1,1)} \to I_{n-2}$ has 1-dim kernel $\mathbb{Q}(q)v_2$ since $(T_1{+}1)v_2 = 0$.

Set $z := R''_{n-1}(w) \in I_{n-2}$. We claim that the matrix of R''_{n-1} from the source basis $(v_3, v_4, \ldots, v_n)$ (in the quotient $V_{(n-1,1)}/\mathbb{Q}(q)v_2$) to the image basis $(v_2, v_3, \ldots, v_{n-3}, u_{n-2}, v_n)$ is lower-triangular with constant nonzero diagonal $(1+q)^{n-3}$.

Computation of the matrix. Let $L_j := (T_j{+}1)(T_{j-1}{+}1)\cdots(T_1{+}1)$. We track $L_j v_l$ for $l \geq 3$ and $j = 1, 2, \ldots, n-2$. By induction on j,

$$L_j v_l \;=\; \begin{cases} (1+q)^j q_{l-1}\big(v_{l-1} + B_{l,l}v_l + B_{l,l+1}v_{l+1} + \cdots + B_{l,j-1}v_{j-1} + B_{l,j}u_j\big) & \text{if } 3 \leq l \leq j+1, \\ (1+q)^j v_l & \text{if } l \geq j+2, \end{cases} \tag{2}$$

where $B_{l,k} := \prod_{i=l-1}^{k-1} \beta_i p_{i+1} = \prod_{i=l-1}^{k-1} \gamma_i$ (empty product equals 1).

The base $j = 1$ gives $L_1 v_l = (1+q)v_l$ for $l \geq 3$. For the inductive step $j-1 \to j$: the operator $T_j{+}1$ acts as $(1+q)$ on v_k for $k \neq j, j+1$, and on the block $\mathrm{span}(v_j, v_{j+1})$ as

$$(T_j{+}1)v_j = (1+q)p_j u_j, \qquad (T_j{+}1)v_{j+1} = u_j = v_j + \beta_j v_{j+1},$$

using $q_j = 1/(1+q)$. Thus $(T_j{+}1)u_{j-1} = (T_j{+}1)(v_{j-1} + \beta_{j-1}v_j) = (1+q)v_{j-1} + (1+q)\beta_{j-1}p_j u_j = (1+q)(v_{j-1} + \gamma_{j-1}u_j)$. Substituting these into the inductive form for L_{j-1} and collecting yields (2).

Reading off the matrix. Apply at $j = n-2$. For source v_l with $3 \leq l \leq n-1$, the image $L_{n-2}v_l$ is supported on the image-basis vectors $v_{l-1}, v_l, \ldots, v_{n-3}, u_{n-2}$, with leading coefficient $(1+q)^{n-2}q_{l-1} = (1+q)^{n-3}$ (using $q_{l-1} = 1/(1+q)$) at v_{l-1}. For $l = n-1$: $L_{n-2}v_{n-1} = (1+q)^{n-2}q_{n-2}u_{n-2} = (1+q)^{n-3}u_{n-2}$. For $l = n$: $L_{n-2}v_n = (1+q)^{n-2}v_n$.

In short, the matrix M from source basis $(v_3, v_4, \ldots, v_n)$ to image basis $(v_2, v_3, \ldots, v_{n-3}, u_{n-2}, v_n)$ has:

- Column for v_l, $3 \leq l \leq n-1$: leading $(1+q)^{n-3}$ at row v_{l-1} (or row u_{n-2} if $l = n-1$); off-diagonal entries $(1+q)^{n-3}\prod_{i=l-1}^{k-1}\gamma_i$ at row v_k (or u_{n-2} for $k = n-2$), for $l \leq k \leq n-2$.

- Column for v_n: $(1+q)^{n-2}$ at row v_n, zero elsewhere.

Reading column-by-column, this is lower-triangular with all diagonal entries equal to $(1+q)^{n-3}$ in the rows above v_n, and $(1+q)^{n-2}$ on the v_n-diagonal.

Back-substitution. Solve $z = Mw$ row by row. Index the source basis as $w_3, w_4, \ldots, w_n$ and the image basis as $z^{(v_2)}, z^{(v_3)}, \ldots, z^{(v_{n-3})}, z^{(u_{n-2})}, z^{(v_n)}$.

Row v_2: $z^{(v_2)} = (1+q)^{n-3} w_3$, so $w_3 = z^{(v_2)}/(1+q)^{n-3}$.

Row v_k, $3 \le k \le n-3$: $z^{(v_k)} = (1+q)^{n-3}\Big[\sum_{l=3}^{k} \Pi_{l,k}\, w_l \;+\; w_{k+1}\Big]$, where $\Pi_{l,k} := \prod_{i=l-1}^{k-1} \gamma_i$.

We claim by induction on k that

$$w_{k+1} \;=\; \frac{z^{(v_k)} - \gamma_{k-1}\, z^{(v_{k-1})}}{(1+q)^{n-3}} \tag{3}$$

for $3 \le k \le n-3$ (with $z^{(v_2)}$ taking the place of $z^{(v_{k-1})}$ at $k=3$). The case $k=3$: $z^{(v_3)} = (1+q)^{n-3}[\gamma_2 w_3 + w_4]$, so $w_4 = z^{(v_3)}/(1+q)^{n-3} - \gamma_2 w_3 = (z^{(v_3)} - \gamma_2 z^{(v_2)})/(1+q)^{n-3}$, matching (3).

For the inductive step $k-1 \to k$: by the inductive hypothesis, $w_{l+1} = (z^{(v_l)} - \gamma_{l-1} z^{(v_{l-1})})/(1+q)^{n-3}$ for $l < k$. Substituting into the row-v_k equation and using the telescoping identity

$$\Pi_{l+1,k} \;=\; \gamma_l\, \Pi_{l+2,k} \qquad \Big(\text{since } \Pi_{l+1,k} = \prod_{i=l}^{k-1} \gamma_i,\ \Pi_{l+2,k} = \prod_{i=l+1}^{k-1} \gamma_i\Big), \tag{4}$$

all terms with $z^{(v_l)}$ for $l \le k-2$ cancel, leaving exactly (3). Specifically, after substituting, the contribution of $z^{(v_l)}$ to w_{k+1} comes from two sources: the direct $-\Pi_{l+1,k} z^{(v_l)}$ term (from substituting w_{l+1} via the inductive formula and dividing by $M_{k,k}$), and the indirect $+\Pi_{l+2,k}\gamma_l z^{(v_l)}$ term (from the $\gamma_l z^{(v_l)}$ part of w_{l+2} via the inductive formula). By (**??**), $\Pi_{l+2,k}\gamma_l = \Pi_{l+1,k}$, so

$$\Big[-\Pi_{l+1,k} + \Pi_{l+2,k}\gamma_l\Big] \cdot \frac{z^{(v_l)}}{(1+q)^{n-3}} \;=\; 0$$

for $2 \le l \le k-2$. The $z^{(v_{k-1})}$ contribution survives as $-\gamma_{k-1}\, z^{(v_{k-1})}/(1+q)^{n-3}$ (no offsetting indirect term).

Row u_{n-2}: $z^{(u_{n-2})} = (1+q)^{n-3}\big[\sum_{l=3}^{n-2} \Pi_{l,n-2} w_l + w_{n-1}\big]$. By an identical telescoping argument, all $z^{(v_l)}$ contributions for $l \le n-4$ cancel, leaving

$$w_{n-1} \;=\; \frac{z^{(u_{n-2})} - \gamma_{n-3}\, z^{(v_{n-3})}}{(1+q)^{n-3}}. \tag{5}$$

Translating to y. The element $y \in \operatorname{span}(v_2, \ldots, v_{n-2})$ relates to z via the constraint $(T_{n-1}{+}1)z = y$. Since y has no v_{n-1} or v_n component, and $T_{n-1}{+}1$ has eigenvalues $1+q$ on $\operatorname{span}(v_2, \ldots, v_{n-2})$ and 0 on $\operatorname{span}(u'_{n-1})$:

$$z \;=\; \tfrac{1}{1+q} y \;+\; \mu\, u'_{n-1}$$

for some $\mu \in \mathbb{Q}(q)$ chosen so that $z \in I_{n-2}$. Working out the constraint (May-8 evening, eq. (2.2)) yields

$$z^{(v_l)} = y_{v_l}/(1+q), \quad 2 \le l \le n-3, \qquad z^{(u_{n-2})} = y_{v_{n-2}}/((1+q)\alpha_{n-2}) = y_{v_{n-2}}/(1+q),$$

using $\alpha_{n-2} = 1$.

Substituting into (3) and (4): for $3 \le k \le n-3$,

$$w_{k+1} = \frac{y_{v_k}/(1+q) - \gamma_{k-1} y_{v_{k-1}}/(1+q)}{(1+q)^{n-3}} = \frac{y_{v_k} - \gamma_{k-1}\, y_{v_{k-1}}}{(1+q)^{n-2}};$$

and for w_{n-1}:

$$w_{n-1} = \frac{y_{v_{n-2}}/(1+q) - \gamma_{n-3} y_{v_{n-3}}/(1+q)}{(1+q)^{n-3}} = \frac{y_{v_{n-2}} - \gamma_{n-3}\, y_{v_{n-3}}}{(1+q)^{n-2}}.$$

Setting $l = k+1$ in the first and matching the second: (1) is established for all $3 \le l \le n-1$.

The case $l = n$ (Image Lemma) was proved May-8 evening. $\square$

Remark 2. The Strong Image formula (1) extends the Image Lemma from w_{v_n} (which depends only on $y_{v_{n-2}}$) to every "intermediate" coordinate w_{v_l} for $l = 3, \dots, n-1$, each of which depends on *exactly two* consecutive coordinates of y.

This local structure is the key technical ingredient in what follows: it converts the question of (Q_m) on K_m into a linear-functional non-vanishing question on K_{m-1}.

3 Slopes on the new generators

We now use the Strong Image formula to extract two "slopes" on the new-generator quotient of K_l, indexed by parity.

Definition 3 (Slopes μ_l and ν_l)**.** For $l \ge 4$, let $\xi^{(l)} \in K_l$ be any vector with $\xi^{(l)} \notin K_{l-2}^{\text{embed}}$ (such a vector exists since K_{l-2}^{embed} has codimension 1 in K_l).

- For l **odd** ($l \ge 5$): $\xi^{(l)}_{v_l} = 0$ by (P_l) and $\xi^{(l)}_{v_{l-1}} \ne 0$ by (Q_l) (the latter is automatic from $(P_l), (P_{l-1})$ as shown in May-9 §2). Define

$$\mu_l \;:=\; \frac{\xi^{(l)}_{v_{l-1}}}{\xi^{(l)}_{v_{l-2}}} \quad (\text{when } \xi^{(l)}_{v_{l-2}} \ne 0).$$

- For l **even** ($l \ge 4$): $\xi^{(l)}_{v_l} \ne 0$ by (P_l). Define

$$\nu_l \;:=\; \frac{\xi^{(l)}_{v_{l-1}}}{\xi^{(l)}_{v_l}}.$$

Both are independent of the choice of $\xi^{(l)}$ within its coset modulo K_{l-2}^{embed} (since $K_{l-2}^{\text{embed}} \subset \operatorname{span}(v_2, \dots, v_{l-2})$ has zero v_{l-1}, v_l components).

Lemma 4 (Base values)**.** $\nu_4 = -1/\gamma_2$ *and* $\mu_5 = -\gamma_2$.

Proof. $K_4 = \operatorname{span}(v_2, v_3 - cv_4)$ with $c = \gamma_2$ (verified in May-9 nesting paper, via $c = q_2\beta_2 p_3/q_3$). The new generator $\eta^{(4)} = v_3 - \gamma_2 v_4$ gives $\nu_4 = 1/(-\gamma_2) = -1/\gamma_2$.

For μ_5: $K_5 = \operatorname{span}(v_2, v_3 - cv_4)$ as well (since (P_5) for 5 odd gives $v_5^*(K_5) = 0$, and the dim and $K_4^{\text{embed}} \subset K_5$ force equality). So $\xi^{(5)} = v_3 - \gamma_2 v_4$ (taken modulo $\mathbb{Q}(q)v_2 = K_3^{\text{embed}}$), giving $\mu_5 = -\gamma_2/1 = -\gamma_2$. $\square$

Lemma 5 (Recursion for μ_l, l odd ≥ 7)**.** *Assume (R_{l-1}) holds at level $l-1$ even. Then*

$$\mu_l \;=\; \frac{-\gamma_{l-3}}{1-\gamma_{l-4}\,\nu_{l-3}}. \tag{6}$$

Proof. Pick $\xi^{(l)} \in K_l$ new generator. By the May-9 reduction analysis at level l: $R_l'(\xi^{(l)}) = y$ lies in $D_l \cap K_{l-1}^{\text{embed}} = K_{l-1}^{\text{embed}} \cap \operatorname{span}(v_2, \ldots, v_{l-2})$. By (P_{l-1}) for $l-1$ even, $v_{l-1}^*(K_{l-1}) \neq 0$ but we additionally need to know the structure of $K_{l-1} \cap \{v_{l-1}^* = 0\}$.

By (R_{l-1}) assumed: v_{l-2}^* and v_{l-1}^* are proportional on K_{l-1} with nonzero ratio; hence $K_{l-1} \cap \{v_{l-1}^* = 0\} = K_{l-1} \cap \{v_{l-2}^* = 0\}$, and by nesting + dimension this equals K_{l-3}^{embed} (in $V_{(l-2,1)}$).

So $y \in K_{l-3}^{\text{embed}}$, which has support on $v_2, \ldots, v_{l-3}$ since $K_{l-3} \subset V_{(l-4,1)}$ (natural embedding). In particular, $y_{v_{l-3}}$ may be nonzero while $y_{v_{l-2}} = 0$.

By Strong Image at level $n = l$:

$$\begin{aligned}
\xi^{(l)}_{v_{l-1}} &= \frac{y_{v_{l-2}} - \gamma_{l-3} y_{v_{l-3}}}{(1+q)^{l-2}} \;=\; \frac{-\gamma_{l-3}\, y_{v_{l-3}}}{(1+q)^{l-2}},\\
\xi^{(l)}_{v_{l-2}} &= \frac{y_{v_{l-3}} - \gamma_{l-4}\, y_{v_{l-4}}}{(1+q)^{l-2}}.
\end{aligned}$$

Hence

$$\mu_l \;=\; \frac{\xi^{(l)}_{v_{l-1}}}{\xi^{(l)}_{v_{l-2}}} \;=\; \frac{-\gamma_{l-3}\, y_{v_{l-3}}}{y_{v_{l-3}} - \gamma_{l-4}\, y_{v_{l-4}}} \;=\; \frac{-\gamma_{l-3}}{1-\gamma_{l-4}\,(y_{v_{l-4}}/y_{v_{l-3}})}.$$

Finally the ratio $y_{v_{l-4}}/y_{v_{l-3}}$ on K_{l-3}^{embed}: since $K_{l-3} = K_{l-5}^{\text{embed}} \oplus \mathbb{Q}(q)\eta^{(l-3)}$ and $K_{l-5}^{\text{embed}} \subset \operatorname{span}(v_2, \ldots, v_{l-5})$ (both v_{l-4}, v_{l-3} coordinates vanish on K_{l-5}^{embed}), the ratio equals $\eta^{(l-3)}_{v_{l-4}}/\eta^{(l-3)}_{v_{l-3}} = \nu_{l-3}$. Substituting gives (5). $\square$

Lemma 6 (Recursion for ν_l, l even ≥ 6)**.**

$$\nu_l \;=\; \frac{1-\gamma_{l-3}/\mu_{l-1}}{(1+q)^{l-2}\, C_l}, \qquad C_l \;=\; \frac{\beta_{l-2}\beta_{l-1}'}{(1+q)^{l-1}}. \tag{7}$$

Proof. Pick $\eta^{(l)} \in K_l$ new generator. Then $R_l'(\eta^{(l)}) = z \in K_{l-1}^{\text{embed}}$ where K_{l-1} is at level $l-1$ odd, so z has support on $\operatorname{span}(v_2, \ldots, v_{l-2})$ (by (P_{l-1})). By Image Lemma: $\eta^{(l)}_{v_l} = C_l z_{v_{l-2}}$. Since $\eta^{(l)} \notin K_{l-2}^{\text{embed}}$ implies $\eta^{(l)}_{v_l} \neq 0$, we have $z_{v_{l-2}} \neq 0$.

By Strong Image: $\eta^{(l)}_{v_{l-1}} = (z_{v_{l-2}} - \gamma_{l-3} z_{v_{l-3}})/(1+q)^{l-2}$. So

$$\nu_l = \frac{\eta^{(l)}_{v_{l-1}}}{\eta^{(l)}_{v_l}} = \frac{z_{v_{l-2}} - \gamma_{l-3} z_{v_{l-3}}}{(1+q)^{l-2}\, C_l\, z_{v_{l-2}}} = \frac{1-\gamma_{l-3}\,(z_{v_{l-3}}/z_{v_{l-2}})}{(1+q)^{l-2}\, C_l}.$$

The ratio $z_{v_{l-3}}/z_{v_{l-2}}$ on K_{l-1} (odd): since $K_{l-1} = K_{l-3}^{\text{embed}} \oplus \mathbb{Q}(q)\xi^{(l-1)}$ and $K_{l-3}^{\text{embed}} \subset \operatorname{span}(v_2, \ldots, v_{l-4})$ (the latter inclusion since $K_{l-3} \subset V_{(l-4,1)}$ embeds as $\operatorname{span}(v_2, \ldots, v_{l-4})$ in $V_{(l-2,1)}$, with no v_{l-3}, v_{l-2} components), the ratio is determined by $\xi^{(l-1)}$: $z_{v_{l-3}}/z_{v_{l-2}} = \xi^{(l-1)}_{v_{l-3}}/\xi^{(l-1)}_{v_{l-2}} = 1/\mu_{l-1}$.

Substituting yields (6). $\square$

4 Sign induction

We now prove an alternating sign property of the slopes at any positive real q, treating $q > 0$ as a real variable in $\mathbb{R}_{>0}$.

Lemma 7 (Positivity of Hoefsmit constants)**.** *At any real $q > 0$, the constants $\beta_i, p_i, q_i, \gamma_i$ are strictly positive, and β_i' is strictly negative. In particular, $C_l = \beta_{l-2}\beta_{l-1}'/(1+q)^{l-1} < 0$.*

Proof. $[k] = (q^k - 1)/(q-1) > 0$ at $q > 0, q \neq 1$, and $[k]|_{q=1} = k > 0$. Therefore $\beta_i = q[i-1]/[i] > 0$, $p_i = [i+1]/([2][i]) > 0$, $q_i = 1/[2] > 0$, and $\gamma_i = \beta_i p_{i+1} > 0$. $\beta_i' = -[i+1]/[i] < 0$. Hence $C_l = (\text{positive}) \cdot (\text{negative})/(\text{positive}) < 0$. □

Theorem 8 (Alternating sign)**.** *For all real $q > 0$:*

- *$\nu_l < 0$ for every even $l \geq 4$.*
- *$\mu_l < 0$ for every odd $l \geq 5$.*

The proof is by strong induction simultaneously establishing $(P_l), (Q_l), (R_l)$ along with these signs.

Proof. Order the levels $4 < 5 < 6 < 7 < \cdots$ and induct on l.

Bases.

- Level 3: $K_3 = \mathbb{Q}(q)v_2$, $(P_3), (Q_3)$ trivial.
- Level 4: $K_4 = \mathrm{span}(v_2, v_3 - \gamma_2 v_4)$ (computed). $(P_4), (Q_4), (R_4)$ all hold. $\nu_4 = -1/\gamma_2 < 0$ by Lemma 7.
- Level 5: $K_5 = \mathrm{span}(v_2, v_3 - \gamma_2 v_4)$ (extended by zero on v_5). $(P_5), (Q_5)$ hold, $\mu_5 = -\gamma_2 < 0$.

Inductive step. Suppose all properties (sign, $(P), (Q), (R)$) hold at all levels $< l$.

Case l odd, $l \geq 7$. By induction, (R_{l-1}) holds (since $l-1$ even is $< l$, and (R_{l-1}) follows from $(P_{l-1}), (Q_{l-1})$ via May-9 Reduction Theorem). Apply Lemma 5: $\mu_l = -\gamma_{l-3}/(1 - \gamma_{l-4}\nu_{l-3})$.

By induction, $\nu_{l-3} < 0$ (since $l-3$ even is $< l$). By Lemma 7, $\gamma_{l-4} > 0$. Hence $\gamma_{l-4}\nu_{l-3} < 0$, so $1 - \gamma_{l-4}\nu_{l-3} > 1 > 0$. Therefore

$$\mu_l \;=\; \frac{-\gamma_{l-3}}{\text{positive}} \;<\; 0.$$

We need to check that $\xi^{(l)}_{v_{l-2}} \neq 0$ (so that μ_l is finite). By Lemma 5's computation, $\xi^{(l)}_{v_{l-2}} = (y_{v_{l-3}}/(1+q)^{l-2})(1 - \gamma_{l-4}\nu_{l-3})$, which is nonzero since both factors are.

Properties $(P_l), (Q_l)$ (with (Q_l) for l odd automatic): established via the May-8 evening induction from $(P_{l'}), (R_{l'})$ at smaller l'.

Case l even, $l \geq 6$. By induction, $\mu_{l-1} < 0$ (since $l-1$ odd is $< l$, with $(P_{l-1}), (Q_{l-1})$ valid). Apply Lemma 6: $\nu_l = (1 - \gamma_{l-3}/\mu_{l-1})/((1+q)^{l-2}C_l)$.

By induction, $\mu_{l-1} < 0$, $\gamma_{l-3} > 0$, hence $\gamma_{l-3}/\mu_{l-1} < 0$, so $1 - \gamma_{l-3}/\mu_{l-1} > 1 > 0$. By Lemma 7, $C_l < 0$, so $(1+q)^{l-2}C_l < 0$. Therefore

$$\nu_l \;=\; \frac{\text{positive}}{\text{negative}} \;<\; 0.$$

For (P_l) at level l even: by May-8 evening induction, $(P_l) \Leftarrow (P_{l-1}), (Q_{l-1})$ (which we have).

For (Q_l) at level l even: see Theorem 9 below.

For (R_l) at level l even: by May-9 Reduction Theorem, $(R_l) \Leftarrow (P_l), (Q_l)$, Nesting, all of which we have. □

5 The main theorem

Theorem 9 (Property (Q_m) for all even m)**.** *For every even $m \geq 4$, the linear functional v_{m-1}^* is not identically zero on K_m. Equivalently, there exists $w \in K_m$ with $w_{v_{m-1}} \neq 0$.*

Proof. The case $m = 4$ is base: $K_4 = \mathrm{span}(v_2, v_3 - \gamma_2 v_4)$ contains $v_3 - \gamma_2 v_4$ with v_3-coordinate $1 \neq 0$.

For $m \geq 6$ even: by Theorem 8 (sign induction established at all levels $< m$), $\mu_{m-1} < 0$ at every $q > 0$. By Lemma 7, $\gamma_{m-3} > 0$. Therefore

$$\mu_{m-1} \; - \; \gamma_{m-3} \; < \; 0$$

at every $q > 0$. Since $\mu_{m-1} - \gamma_{m-3}$ is a rational function in q that is strictly negative at every positive real q, it cannot be identically zero. Hence $\mu_{m-1} - \gamma_{m-3} \neq 0$ in $\mathbb{Q}(q)$.

Pick $\xi^{(m-1)} \in K_{m-1}$ a new generator of K_{m-1} with $\xi^{(m-1)}_{v_{m-2}} \neq 0$ (exists by (Q_{m-1}), automatic for $m-1$ odd). By Theorem 8's argument, $\xi^{(m-1)}_{v_{m-3}} \neq 0$ as well. Embed $\xi^{(m-1)}$ into $V_{(m-1,1)}$ via the natural embedding, and consider the linear functional

$$\phi_m \; := \; v_{m-2}^* - \gamma_{m-3}\, v_{m-3}^*$$

applied to $\xi^{(m-1)}$:

$$\phi_m(\xi^{(m-1)}) \; = \; \xi^{(m-1)}_{v_{m-2}} - \gamma_{m-3}\, \xi^{(m-1)}_{v_{m-3}} \; = \; \xi^{(m-1)}_{v_{m-3}} \big(\mu_{m-1} - \gamma_{m-3}\big) \; \neq \; 0.$$

Now lift $\xi^{(m-1)}$ to $w \in K_m$ via $R_m'^{-1}$: pick any $w \in V_{(m-1,1)}$ with $R_m'(w) = \xi^{(m-1)}$. By the May-8 evening Reduction Theorem (using the Image Lemma and K_{m-1} analysis), such w lies in K_m provided $\xi^{(m-1)} \in D_m \cap K_{m-1}^{\mathrm{embed}}$. Since $\xi^{(m-1)}_{v_{m-1}} = 0$ ((P_{m-1}) for $m-1$ odd), $\xi^{(m-1)} \in \mathrm{span}(v_2, \ldots, v_{m-2}) \subset D_m$. And by the inductive hypothesis $\xi^{(m-1)} \in K_{m-1}^{\mathrm{embed}}$. So $w \in K_m$.

By Strong Image at level $n = m$ applied to w with $y = \xi^{(m-1)}$:

$$w_{v_{m-1}} \; = \; \frac{\xi^{(m-1)}_{v_{m-2}} - \gamma_{m-3}\, \xi^{(m-1)}_{v_{m-3}}}{(1+q)^{m-2}} \; = \; \frac{\phi_m(\xi^{(m-1)})}{(1+q)^{m-2}} \; \neq \; 0.$$

This proves (Q_m). ☐

Corollary 10 (Hook rank formula, unconditional)**.** *For every $n \geq 2$,*

$$\mathrm{rank}\, \Pi^{S_n}|_{V_{(n-1,1)}} \; = \; \lceil \tfrac{n-2}{2} \rceil.$$

Proof. By Theorem 9, (Q_m) holds for every even $m \geq 4$. Combined with the May-9 Reduction Theorem and the rank-formula proof in May-9 §4, the rank formula holds for all $n \geq 2$. ☐

6 Computational verification

The Strong Image Coefficient Formula (1) is verified symbolically at $n = 4, 5, 6, 7, 8$ in `verify_strong_image.py`, where the formula is matched against direct linear-algebra computation of w given y.

The slopes μ_l, ν_l have been verified explicitly:

- $\nu_4 = -1/\gamma_2 = -(q+1)(q^2+q+1)/(q(q^2+1))$.
- $\mu_5 = -\gamma_2 = -q(q^2+1)/((q+1)(q^2+q+1))$.

- $\nu_6 =$ (computed in May-8 evening Table).
- $\mu_7 = -\gamma_2\gamma_4/(\gamma_2+\gamma_3) = -q(q^2+1)(q^2-q+1)(q^2+q+1)/((q+1)(2q^2-q+2)(q^4+q^3+q^2+q+1))$.

The identity $\mu_{m-1} - \gamma_{m-3}$ for (Q_m) at $m = 6, 8$:

- $m = 6$: $\mu_5 - \gamma_3 = -(\gamma_2 + \gamma_3) = -q(2q^2 - q + 2)/((q + 1)(q^2 + 1))$, nonzero.
- $m = 8$: $\mu_7-\gamma_5 = -q(q^2+1)(3q^4-2q^3+4q^2-2q+3)/((q+1)(q^2-q+1)(q^2+q+1)(2q^2-q+2))$, nonzero.

Both negative at $q > 0$ as predicted by Theorem 8.

7 Honest assessment

What this paper proves.

1. Theorem 1 (Strong Image Coefficient Formula). Rigorous via direct triangular matrix inversion of R''_{n-1} on $V_{(n-1,1)}$.
2. Lemmas 5 and 6: recursive formulas for the slopes μ_l, ν_l.
3. Theorem 8 (alternating sign): both slopes strictly negative at every positive real q.
4. Theorem 9: (Q_m) holds for all even $m \geq 4$.
5. Corollary 10: hook rank formula proved unconditionally.

Where Hoefsmit-specific computation enters. The proof crucially uses positivity of the explicit Hoefsmit constants β_i, p_i, γ_i at $q > 0$, and negativity of β'_i. These hold because $V_{(n-1,1)}$'s seminormal block matrices have a uniform sign structure under our normalization.

Inductive scope. The induction proceeds simultaneously on $(P_l), (Q_l), (R_l)$ for $l \leq m$, with the sign properties of μ_l, ν_l as additional inductive invariants. Each level requires properties at all strictly smaller levels. No level uses its own (R_l) or (Q_l) in deriving the next level.

Files.

- `~/projects/proofs/2026-05-10-Q-m-via-strong-image.tex` (this writeup).
- `~/projects/scratch/2026-05-09-nesting-identity/check_identity.py` (Hoefsmit constant values).
- `~/projects/scratch/2026-05-08-kernel-dichotomy/check_n8_proportionality.py` (computational verification of $(Q_m), (R_m)$ at $m \leq 8$).

What remains. With (Q_m) proved, the May-9 Reduction Theorem closes the hook rank formula unconditionally. Future directions:

- Closed form for the μ_l, ν_l progressions (continued fraction in the Hoefsmit gammas).
- Generalization to other 2-row partitions $(n - r, r)$ for $r \geq 2$.
- Cellular interpretation: does the alternating sign reflect a duality between the seminormal form and another preferred basis (Kazhdan-Lusztig, web bases, etc.)?